```
In[ ]:= (*PARTE 1*)
```

```
In[ ]:= datae1 = {{12, 0.911 - 1}, {13, 0.95 - 1}, {14, 0.92 - 1},
           {15, 0.802 - 1}, {16, 0.91 - 1}, {17, 0.849 - 1}, {18, 0.92 - 1}};
```

```
In[ ]:= dataE1 = {{12, -0.0069}, {13, -0.0007}, {14, -0.004},
           {15, -0.037}, {16, -0.003}, {17, -0.022}, {18, -0.0042}};
```

```
In[ ]:= datap1 = {{12, -0.0132}, {13, -0.003}, {14, -0.0129},
           {15, -0.066}, {16, -0.013}, {17, -0.054}, {18, -0.0134}};
```

```
In[ ]:= pplote1 = ListPlot[datae1 , PlotStyle → {Blend[{Blue, Green}], Thickness[0.003]},
                  representación de lista  estilo de repre···  mezcla···  azul  verde  grosor
           PlotLegends → {"e - 1"}, BaseStyle → {FontFamily → "Calibri", FontSize → 14},
           leyendas de representación  estilo base  familia de tipo de letra  tamaño de tipo de letra
           PlotStyle → {Blend[{Blue}], Thick}, PlotRange → {{11.8, 18.2}, {-0.23, 0.04}},
           estilo de repre···  mezcla···  azul  grueso  rango de representación
           Frame → True, FrameLabel → {Style["n ", Bold, 16], Style[" ", 16, Bold]}];
           marco  verd···  etiqueta de marco  estilo  negrita  estilo  negrita
```

··· Blend: {■} is not a valid list of colors or images, or pairs of a real number and a color or an image.

```
In[ ]:= pplotE1 =
         ListPlot[dataE1 , PlotStyle → {Black, Thickness[0.003]}, PlotLegends → {"ΔE / J"}];
         representación de lista  estilo de repre···  negro  grosor  leyendas de representación
```

```
In[ ]:= pplotp1 = ListPlot[datap1 ,
                  representación de lista
           PlotStyle → {Red, Thickness[0.003]}, PlotLegends → {"Δp / kg m s⁻¹"}];
           estilo de repre···  rojo  grosor  leyendas de representación
```

```
In[ ]:= Needs["ErrorBarPlots`"];
        necesita
```

··· General: ErrorBarPlots` is now obsolete. The legacy version being loaded may conflict with current functionality. See the Compatibility Guide for updating information.

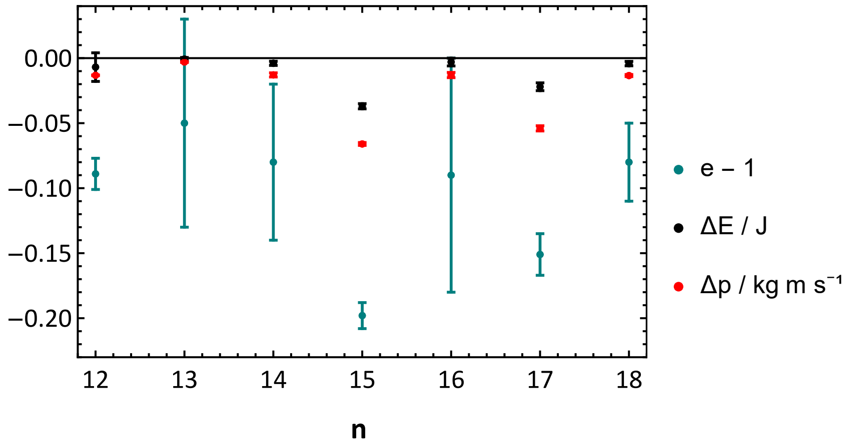
```
In[ ]:= dataerre1 = ErrorListPlot[{{{12, 0.911 - 1}, ErrorBar[-0.012, 0.012]},
            {{13, 0.95 - 1}, ErrorBar[-0.08, 0.08]}, {{14, 0.92 - 1}, ErrorBar[-0.06, 0.06]},
            {{15, 0.802 - 1}, ErrorBar[-0.01, 0.01]}, {{16, 0.91 - 1}, ErrorBar[-0.09, 0.09]},
            {{17, 0.849 - 1}, ErrorBar[-0.016, 0.016]}, {{18, 0.92 - 1}, ErrorBar[-0.03, 0.03]}},
           AxesOrigin → {0, 0}, PlotStyle → {Blend[{Blue, Green}], Thickness[0.005]}];
           origen de ejes  estilo de repre···  mezcla···  azul  verde  grosor
```

```
In[ ]:= dataerrE1 = ErrorListPlot[{{{12, -0.0069}, ErrorBar[-0.0011, 0.011]},
            {{13, -0.0007}, ErrorBar[-0.0012, 0.0012]},
            {{14, -0.004}, ErrorBar[-0.0016, 0.0016]}, {{15, -0.037}, ErrorBar[-0.002, 0.002]},
            {{16, -0.003}, ErrorBar[-0.003, 0.003]}, {{17, -0.022}, ErrorBar[-0.003, 0.003]},
            {{18, -0.0042}, ErrorBar[-0.0017, 0.0017]}},
           AxesOrigin → {0, 0}, PlotStyle → {Black, Thickness[0.005]}];
           origen de ejes  estilo de repre···  negro  grosor
```

In[ ]:=
```
dataerrp1 = ErrorListPlot[{{{12, -0.0132}, ErrorBar[-0.0003, 0.0003]},
    {{13, -0.003}, ErrorBar[-0.0003, 0.0003]}, {{14, -0.0129},
     ErrorBar[-0.0016, 0.0016]}, {{15, -0.066}, ErrorBar[-0.0011, 0.0011]},
    {{16, -0.013}, ErrorBar[-0.002, 0.002]}, {{17, -0.054}, ErrorBar[-0.002, 0.002]},
    {{18, -0.0134}, ErrorBar[-0.0008, 0.0008]}},
   AxesOrigin → {0, 0}, PlotStyle → {Red, Thickness[0.005]}];
   origen de ejes          estilo de repre··· rojo  grosor
```

In[ ]:=
```
Show[pplote1, pplotE1, pplotp1, dataerre1, dataerrE1, dataerrp1]
muestra
```

Out[ ]=



In[ ]:=
```
(*PARTE 2*)
```

In[ ]:=
```
dataerre2 = ErrorListPlot[{{{1, 0.99 - 1}, ErrorBar[-0.04, 0.04]},
    {{2, 0.963 - 1}, ErrorBar[-0.012, 0.012]}, {{3, 0.99 - 1}, ErrorBar[-0.04, 0.04]},
    {{4, 1.05 - 1}, ErrorBar[-0.12, 0.12]}, {{5, 0.973 - 1}, ErrorBar[-0.03, 0.03]},
    {{6, 0.93 - 1}, ErrorBar[-0.04, 0.04]}, {{7, 0.94 - 1}, ErrorBar[-0.03, 0.03]},
    {{8, 0.99 - 1}, ErrorBar[-0.03, 0.03]}}, AxesOrigin → {0, 0},
                                              origen de ejes
   PlotStyle → {Blend[{Blue, Green}], Thickness[0.005]}, PlotLegends → {"e - 1"}];
   estilo de repre··· mezcla··· azul  verde  grosor            leyendas de representación
```

In[ ]:=
```
dataerrE2 = ErrorListPlot[{{{1, -0.0017}, ErrorBar[-0.0013, 0.0013]},
    {{2, -0.0063}, ErrorBar[-0.0015, 0.0015]}, {{3, -0.003}, ErrorBar[-0.003, 0.003]},
    {{4, 0.0002}, ErrorBar[-0.011, 0.011]}, {{5, -0.0022}, ErrorBar[-0.0015, 0.0015]},
    {{6, -0.015}, ErrorBar[-0.012, 0.012]}, {{7, -0.009}, ErrorBar[-0.004, 0.004]},
    {{8, -0.0048}, ErrorBar[-0.0011, 0.0011]}}, AxesOrigin → {0, 0},
                                                  origen de ejes
   PlotStyle → {Black, Thickness[0.005]}, PlotLegends → {"ΔE / J"}];
   estilo de repre··· negro  grosor              leyendas de representación
```

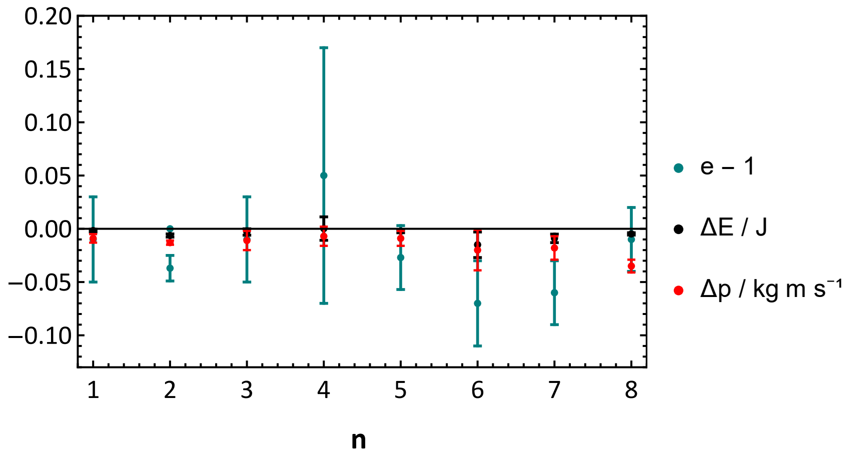
In[ ]:=
```
dataerrp2 = ErrorListPlot[{{{1, -0.009}, ErrorBar[-0.004, 0.004]},
    {{2, -0.013}, ErrorBar[-0.002, 0.002]}, {{3, -0.011}, ErrorBar[-0.009, 0.009]},
    {{4, -0.007}, ErrorBar[-0.009, 0.009]}, {{5, -0.009}, ErrorBar[-0.007, 0.007]},
    {{6, -0.02}, ErrorBar[-0.019, 0.019]}, {{7, -0.018}, ErrorBar[-0.011, 0.011]},
    {{8, -0.035}, ErrorBar[-0.006, 0.006]}}, AxesOrigin → {0, 0},
                                              origen de ejes
   PlotStyle → {Red, Thickness[0.004]}, PlotLegends → {"Δp / kg m s⁻¹"}];
   estilo de repre··· rojo  grosor              leyendas de representación
```

In[ ]:=
```
graph = ListPlot[{-10, 0}, PlotStyle → {Blend[{Blue, Green}], Thickness[0.003]},
        representación de lista   estilo de repre···  mezcla··· azul   verde      grosor
    BaseStyle → {FontFamily → "Calibri", FontSize → 14},
    estilo base       familia de tipo de letra          tamaño de tipo de letra
    PlotStyle → {Blend[{Blue}], Thick}, PlotRange → {{0.8, 8.2}, {-0.13, 0.20}},
    estilo de repre··· mezcla··· azul  grueso    rango de representación
    Frame → True, FrameLabel → {Style["n ", Bold, 16], Style[" ", 16, Bold]}];
    marco    verd··· etiqueta de marco estilo   negrita     estilo          negrita
```

••• Blend: {■} is not a valid list of colors or images, or pairs of a real number and a color or an image.

In[ ]:=
```
Show[graph, dataerre2, dataerrE2, dataerrp2]
muestra
```

Out[ ]=



In[ ]:=
```
(*PARTE 3*)
```

In[ ]:=
```
graph2 = ListPlot[{-10, 0}, PlotStyle → {Blend[{Blue, Green}], Thickness[0.003]},
         representación de lista   estilo de repre···  mezcla··· azul   verde      grosor
    BaseStyle → {FontFamily → "Calibri", FontSize → 14},
    estilo base       familia de tipo de letra          tamaño de tipo de letra
    PlotStyle → {Blend[{Blue}], Thick}, PlotRange → {{0.8, 5.2}, {-0.03, 0.03}},
    estilo de repre··· mezcla··· azul  grueso    rango de representación
    Frame → True, FrameLabel → {Style["n ", Bold, 16], Style[" ", 16, Bold]}];
    marco    verd··· etiqueta de marco estilo   negrita     estilo          negrita
```

••• Blend: {■} is not a valid list of colors or images, or pairs of a real number and a color or an image.

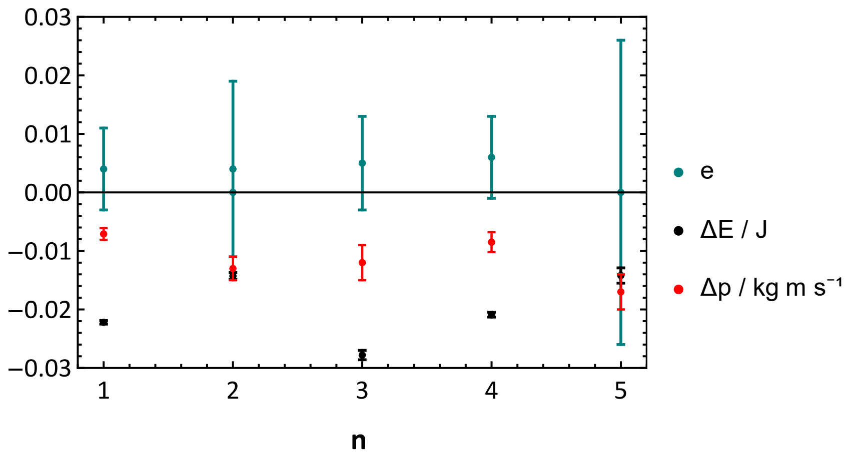
In[ ]:=
```
dataerre3 = ErrorListPlot[
    {{{1, 0.004}, ErrorBar[-0.007, 0.007]}, {{2, 0.004}, ErrorBar[-0.015, 0.015]},
     {{3, 0.005}, ErrorBar[-0.008, 0.008]}, {{4, 0.006}, ErrorBar[-0.007, 0.007]},
     {{5, 0}, ErrorBar[-0.026, 0.026]}}, AxesOrigin → {0, 0},
                                         origen de ejes
    PlotStyle → {Blend[{Blue, Green}], Thickness[0.005]}, PlotLegends → {"e"}];
    estilo de repre··· mezcla··· azul  verde    grosor                leyendas de representación
```

In[ ]:=
```
dataerrp3 = ErrorListPlot[
    {{{1, -0.0071}, ErrorBar[-0.0010, 0.0010]}, {{2, -0.013}, ErrorBar[-0.002, 0.002]},
     {{3, -0.012}, ErrorBar[-0.003, 0.003]}, {{4, -0.0085}, ErrorBar[-0.0017, 0.0017]},
     {{5, -0.017}, ErrorBar[-0.003, 0.003]}}, AxesOrigin → {0, 0},
                                              origen de ejes
    PlotStyle → {Red, Thickness[0.004]}, PlotLegends → {"Δp / kg m s⁻¹"}];
    estilo de repre··· rojo  grosor          leyendas de representación
```

In[◦]:=
```
dataerrE3 = ErrorListPlot[{{{1, -0.0222}, ErrorBar[-0.0003, 0.0003]},
    {{2, -0.0143}, ErrorBar[-0.0006, 0.0006]}, {{3, -0.0278},
     ErrorBar[-0.0008, 0.0008]}, {{4, -0.0209}, ErrorBar[-0.0004, 0.0004]},
    {{5, -0.0142}, ErrorBar[-0.0013, 0.0013]}}, AxesOrigin → {0, 0},
                                                   origen de ejes
   PlotStyle → {Black, Thickness[0.005]}, PlotLegends → {"ΔE / J"}];
   estilo de repre··· negro  grosor             leyendas de representación
```

In[◦]:=
```
Show[graph2, dataerre3, dataerrE3, dataerrp3]
muestra
```

Out[◦]=



In[◦]:=
```
(*MEDIAS*)
```

In[◦]:=
```
tickSpecification = Table[{0 + i}, {i, 1, 3}];
                    tabla
```

In[◦]:=
```
graphmed = ListPlot[{-15, 15}, PlotStyle → {Blend[{Blue, Green}], Thickness[0.003]},
           representación de lista  estilo de repre··· mezcla··· azul  verde  grosor
   BaseStyle → {FontFamily → "Calibri", FontSize → 14}, PlotStyle →
   estilo base  familia de tipo de letra  tamaño de tipo de l··  estilo de representación
    {Blend[{Blue}], Thick}, PlotRange → {{0.8, 3.2}, {-0.12, 0.02}}, Frame → True,
     mezcla··· azul  grueso  rango de representación                   marco  verdadero
   FrameLabel → {Style["Series de datos ", Bold, 16], Style[" ", 16, Bold]},
   etiqueta de marco  estilo  serie             negrita    estilo          negrita
   Ticks → {tickSpecification, Automatic}];
   marcas                      automático
```

••• Blend: {■} is not a valid list of colors or images, or pairs of a real number and a color or an image.

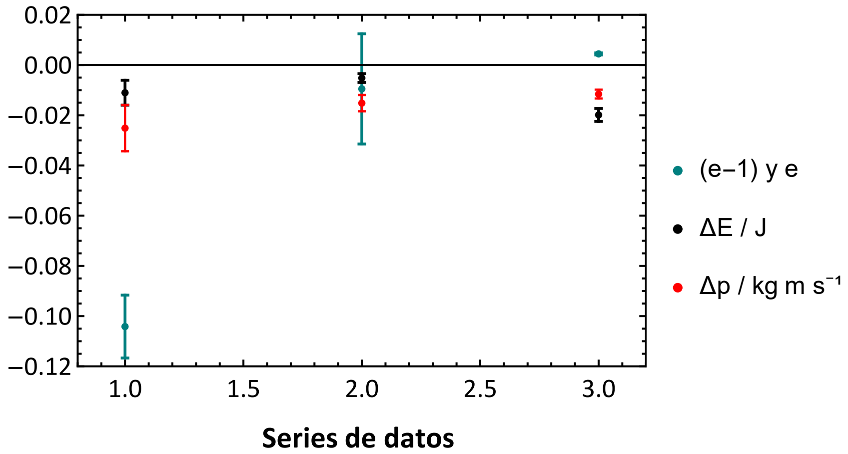
In[◦]:=
```
dataerremed = ErrorListPlot[{{{1, 0.89583 - 1}, ErrorBar[-0.01252, 0.01252]},
    {{2, 0.9905044 - 1}, ErrorBar[-0.021957, 0.021957]},
    {{3, 0.0044543}, ErrorBar[-0.00041, 0.00041]}}, AxesOrigin → {0, 0},
                                                    origen de ejes
   PlotStyle → {Blend[{Blue, Green}], Thickness[0.005]}, PlotLegends → {"(e-1) y e"}];
   estilo de repre··· mezcla··· azul  verde  grosor        leyendas de representación
```

In[◦]:=
```
dataerrpmed = ErrorListPlot[{{{1, -0.02513}, ErrorBar[-0.0092, 0.0092]},
    {{2, -0.01517}, ErrorBar[-0.00324, 0.00324]},
    {{3, -0.011564}, ErrorBar[-0.00176, 0.00176]}}, AxesOrigin → {0, 0},
                                                    origen de ejes
   PlotStyle → {Red, Thickness[0.004]}, PlotLegends → {"Δp / kg m s⁻¹"}];
   estilo de repre··· rojo  grosor           leyendas de representación
```

In[∘]:=
```
dataerrEmed = ErrorListPlot[{{{1, -0.011035}, ErrorBar[-0.00496, 0.00496]},
    {{2, -0.005178}, ErrorBar[-0.00176, 0.00176]},
    {{3, -0.019875}, ErrorBar[-0.00258, 0.00258]}}, AxesOrigin → {0, 0},
                                                   origen de ejes
  PlotStyle → {Black, Thickness[0.005]}, PlotLegends → {"ΔE / J"}];
  estilo de repre⋯ negro grosor                     leyendas de representación
```

In[∘]:=
```
Show[graphmed, dataerremed, dataerrEmed, dataerrpmed]
muestra
```

Out[∘]=



In[∘]:=
```
(*PARTE 1*)
```

In[∘]:=
```
graph4 = ListPlot[{-10, 0}, PlotStyle → {Blend[{Blue, Green}], Thickness[0.003]},
         representación de lista estilo de repre⋯ mezcla⋯ azul verde grosor
  BaseStyle → {FontFamily → "Calibri", FontSize → 14},
  estilo base  familia de tipo de letra  tamaño de tipo de letra
  PlotStyle → {Blend[{Blue}], Thick}, PlotRange → {{0.8, 11.2}, {-0.8, 0.05}},
  estilo de repre⋯ mezcla⋯ azul grueso rango de representación
  Frame → True, FrameLabel → {Style["n ", Bold, 16], Style[" ", 16, Bold]}];
  marco  verd⋯  etiqueta de marco estilo  negrita  estilo  negrita
```

⋯ Blend: {■} is not a valid list of colors or images, or pairs of a real number and a color or an image.

In[∘]:=
```
dataerre4 = ErrorListPlot[{{{1, 0.264 - 1}, ErrorBar[-0.018, 0.018]},
    {{2, 0.6186 - 1}, ErrorBar[-0.0692, 0.0692]},
    {{3, 0.7652 - 1}, ErrorBar[-0.025, 0.025]}, {{4, 0.79649 - 1},
     ErrorBar[-0.018, 0.018]}, {{5, 0.7615 - 1}, ErrorBar[-0.022, 0.022]},
    {{6, 0.242 - 1}, ErrorBar[-0.032, 0.032]}, {{7, 0.8}, ErrorBar[-0.012, 0.012]},
    {{8, 0.81 - 1}, ErrorBar[-0.0334, 0.0334]}, {{9, 0.8 - 1},
     ErrorBar[-0.0335, 0.0335]}, {{10, 0.829 - 1}, ErrorBar[-0.0265, 0.0265]},
    {{11, 0.515 - 1}, ErrorBar[-0.019, 0.019]}}, AxesOrigin → {0, 0},
                                                origen de ejes
  PlotStyle → {Blend[{Blue, Green}], Thickness[0.005]}, PlotLegends → {"e-1"}];
  estilo de repre⋯ mezcla⋯ azul verde grosor        leyendas de representación
```

In[ ]:=
```
dataerrp4 = ErrorListPlot[{{{1, -0.1161}, ErrorBar[-0.0005, 0.0005]},
    {{2, -0.0225}, ErrorBar[-0.00024, 0.00024]}, {{3, -0.0297},
     ErrorBar[-0.0004, 0.0004]}, {{4, -0.0256}, ErrorBar[-0.0003, 0.0003]},
    {{5, -0.0157}, ErrorBar[-0.0001, 0.0001]}, {{6, -0.213},
     ErrorBar[-0.00025, 0.00025]}, {{7, -0.0388}, ErrorBar[-0.00045, 0.00045]},
    {{8, -0.0274}, ErrorBar[-0.00073, 0.00073]}, {{9, -0.0242},
     ErrorBar[-0.0005, 0.0005]}, {{10, -0.026}, ErrorBar[-0.00062, 0.00062]},
    {{11, -0.13777}, ErrorBar[-0.0016, 0.0016]}}, AxesOrigin → {0, 0},
                                                     origen de ejes
   PlotStyle → {Red, Thickness[0.004]}, PlotLegends → {"Δp / kg m s⁻¹"}];
   estilo de repre··· rojo  grosor                leyendas de representación
```

In[ ]:=
```
dataerrE4 = ErrorListPlot[{{{1, -0.042}, ErrorBar[-0.0044, 0.0044]},
    {{2, -0.0039}, ErrorBar[-0.0013, 0.0013]}, {{3, -0.0121},
     ErrorBar[-0.0018, 0.0018]}, {{4, -0.0105}, ErrorBar[-0.00126, 0.00126]},
    {{5, -0.0033}, ErrorBar[-0.000478, 0.000478]}, {{6, -0.135},
     ErrorBar[-0.0061, 0.0061]}, {{7, -0.0244}, ErrorBar[-0.00216, 0.00216]},
    {{8, -0.0068}, ErrorBar[-0.001645, 0.001645]}, {{9, -0.0048},
     ErrorBar[-0.0011, 0.0011]}, {{10, -0.0046}, ErrorBar[-0.00086, 0.00086]},
    {{11, -0.0374}, ErrorBar[-0.0035, 0.0035]}}, AxesOrigin → {0, 0},
                                                   origen de ejes
   PlotStyle → {Black, Thickness[0.005]}, PlotLegends → {"ΔE / J"}];
   estilo de repre··· negro  grosor                leyendas de representación
```

In[ ]:=
```
Show[graph4, dataerre4, dataerrE4, dataerrp4]
muestra
```

Out[ ]=